

Degree-free Holant SI Reading

August 25, 2026

1 Main structure

Assumptions:

- for any $v \in V$, the signature of v is log-concave.
- for $e = uv \in E$, let $\theta_e(i, j) = \lambda_e \frac{f_u(i+1)}{f_u(i)} \frac{f_v(j+1)}{f_v(j)}$ be the increased weight if we add the edge e . We have that $\theta_e(i, j) \in [\gamma, \Theta]$, where γ and Θ are some constants independent of the graph.

Then we have the upper bound of marginal distribution:

$$\Pr_{S \sim \mu} [e \in S] = p_e \leq \frac{\Theta}{1 + \Theta}. \quad (1)$$

Therefore, it is sufficient to show $\text{Cov}(X) \preceq \text{diag}\{\mathbb{E}_\mu[X]\}$.

Let us fix $a \in \mathbb{R}^E$. Define $\phi_e(i, j)$ backwards on $(\{0\} \cup [c_u]) \times (\{0\} \cup [c_v])$ as follows,

$$\phi_e(i, j) = \begin{cases} 0 & , i = c_u \vee j = c_v \\ \theta_e(i, j)(1 - \phi_e(i+1, j+1)) & , \text{o.w.} \end{cases}, \quad (2)$$

and $K_e := \phi_e(d_S(u), d_S(v))$ for $uv = e \in E$. Here ϕ is *intentionally* defined to have the following property:

$$\mathbb{E}_\mu [K_e \mid X_{-e}] = \mathbb{E}_\mu [X_e \mid X_{-e}] \implies \mathbb{E}_\mu [K_e X_f] = \mathbb{E}_\mu [X_e X_f], \forall f \neq e. \quad (3)$$

Define

$$F_a := \sum_{e \in E} a_e X_e \quad \text{and} \quad K_a := \sum_{e \in E} a_e K_e.$$

By [Equation \(3\)](#), we decompose $\text{Var}_\mu[F_a]$ as follows,

$$\begin{aligned} \text{Var}_\mu[F_a] &= \text{Cov}(F_a, K_a) + \mathbb{E}_\mu[F_a^2] - \mathbb{E}_\mu[F_a K_a] \\ &= \text{Cov}(F_a, K_a) + \sum_{e_1, e_2 \in E} a_{e_1} a_{e_2} (\mathbb{E}_\mu[X_{e_1} X_{e_2}] - \mathbb{E}_\mu[X_{e_1} K_{e_2}]) \\ &= \text{Cov}(F_a, K_a) + \sum_{e \in E} a_e^2 (\mathbb{E}_\mu[X_e^2] - \mathbb{E}_\mu[X_e K_e]) \\ &\leq \sqrt{\text{Var}_\mu[F_a] \text{Var}_\mu[K_a]} + \sum_{e \in E} a_e^2 (p_e - \mathbb{E}_\mu[X_e K_e]). \end{aligned}$$

Here the off-diagonal elements cancel and we only need to focus on the diagonal elements.

Proposition 1.1. *There exists some constant $H = H(b, \gamma, \Theta, \lambda)$ such that $|K_e| \leq H$ for $e \in E$.*

Define

$$Q(a) := \sum_{e \in E} a_e^2 p_e = a^\top \text{diag}\{p\} a_e.$$

we have that $\mathbb{E}_\mu [-X_e K_e] \leq \max_{e \in E} |K_e| \Pr_\mu [e \in S] \leq p_e H$. Therefore, by AM-GM inequality,

$$\text{Var}_\mu [F_a] \leq \text{Var}_\mu [K_a] + (1 + H)Q(a). \quad (4)$$

1.1 Upper bound $\text{Var}_\mu [K_a]$

Lemma 1.2. *Consider two Holant measures that agree except at one vertex v . where their signatures are either*

$$(f_v, \mathcal{D}f_v) \quad \text{or} \quad (\delta_{k+1}, \delta_k),$$

*then **log-concavity** implies there is a coupling (S^+, S^-) such that*

$$\sum_{x \in V} |d_{S^+}(x) - d_{S^-}(x)| \leq 2.$$

Let $Y = \mathbf{1}_{c_v - d_S(v) \geq r}$ where c_v is the support length of f_v .

Proposition 1.3. *The above coupling implies the spectral stability of Y .*

Lemma 1.4. *The spectral stability of Y ($\text{Cov}(Y) \preceq C \cdot \text{diag}\{\mathbb{E}[Y]\}$) implies the spectral gap of some resulting kernel. Together with spectral stability, the spectral gap implies the following bound*

$$\text{Var}_\mu [K] \leq 4^b \left(\sum_i \mathbb{E}_\mu [Y_i H_i^2] + \sum_{ij} c_{ij} \mathbb{E}_\mu [Y_i Y_j] \right),$$

where $K(Y) = \sum_{ij} c_{ij} Y_i Y_j$ is the quadratic form of Y and $H_i(Y) = \sum_j c_{ij} Y_j$ is the “first derivative”.

We can also express K_a as the quadratic form of Y variables:

$$K_e = \sum_{e \in E} a_e \sum_{r=1}^{c_u} \sum_{s=1}^{c_v} \alpha_{rs}^e Y_{u,r} Y_{v,s}, \quad (5)$$

where α is “second derivative” of ϕ_e . Define

$$R(a) = \sum_{v \in V} \frac{(\sum_{e: v \in e} p_e a_e)^2}{\mathbb{E}_\mu [Y_{v,1}]}.$$

Plugging [Theorem 1.4](#) into K_e , we have that

$$\begin{aligned} \text{Var}_\mu [K_e] &\lesssim \underbrace{\sum_{v,s=1} \mathbb{E}_\mu [Y_{v,s} H_{v,s}^2]}_{\lesssim Q(a) + R(a)} + \underbrace{\sum_{v,s \geq 2} \mathbb{E}_\mu [Y_{v,s} H_{v,s}^2]}_{\lesssim Q(a)} + \underbrace{\sum_{u,v,s,r} c_{ij} \mathbb{E}_\mu [Y_{u,s} Y_{v,r}]}_{\lesssim Q(a)} \end{aligned}$$

1.2 Hodge decomposition

Let B be the incident matrix. Let h be the vector that solves

$$(W + BPB^\top)h = BP a,$$

where $P = \text{diag}\{p_e\}$ and $W = \text{diag}\{\mathbb{E}_\mu[Y_{v,1}]\}$. Put $z = a - B^\top h$, thereby, $BPz = BP a - BPB^\top h = Wh$. We have that

$$R(a) = (BP a)^\top W^{-1}(BP a), \quad R(z) = h^\top Wh.$$

$$Q(a) = (z + B^\top h)^\top P(z + B^\top h) = Q(z) + R(z) + h^\top BPB^\top h \geq Q(z) + R(z),$$

since $h^\top BPz = h^\top Wh$.

$$F_z = F_a - \sum_{e \in E} (B^\top h)_e X_e \implies \text{Var}_\mu \left[\sum_{e \in E} (B^\top h)_e X_e \right] = \text{Var}_\mu \left[\sum_{v \in E} h_v d_S(v) \right] \underbrace{\lesssim}_{S.S.of Y} R(z)$$

$$\begin{aligned} \text{Var}_\mu[F_a] &\leq \left(\sqrt{\text{Var}_\mu[F_z]} + \sqrt{\text{Var}_\mu[F_{B^\top h}]} \right)^2 \leq 2(\text{Var}_\mu[F_z] + \text{Var}_\mu[F_{B^\top h}]) \\ &\lesssim Q(z) + R(z) \lesssim Q(a). \end{aligned}$$